



THE CHINOOK: Master Bill of Materials (BOM) v2.0

Project: Chinook Axial Flux Motor & StructuralFrame Target: PCBWay Contest & Prototype Production

1. Electronics: Logic & Sensing Board

Component	Specification	Qty	Function
MOSFETs	IRFB3607PBF (75V, 80A)	6	H-Bridge Regen Control
GateDriver	DRV8302	1	3-Phase BLDC Control
ADC	ADS1115(16-bit)	1	SHM (Structural Health) R eader
Instr Amp	AD8226ARMZ	4	Joint Stress Monitoring (St rain Gauges)
WeatherModule	BME280	1	Environmental Data (Vanc ouver Climate)
Resistors	RT0805(0.1%, 100R)	50	Wheatstone Bridge Precisi on
Header	0.5mmpitch, 10-pin	1	Board-to-Board (Logic to Power)

2. Electronics: Power Stator (PCBWay Specs)

Component	Specification	Qty	Function
PowerPCB	130mm, 4oz Copper, TG170	1	Axial Flux PCB Stator
Terminals	XT60-M (Surface Mount)	3	Phase Connections (U, V, W)
Thermal Pad	1mm Silicone (130mm)	1	Stator Heat Dissipation

3. Structural Materials (Bamboo Pole Frame)

Material	Specification	Qty	Function
Bamboo Poles	Structural Grade (Heat Tre ated)	TBD	Main Chassis
Carbon Fiber	300gsmUD Tape	5m	Internal Joint Reinforceme nt
BasaltFiber	200-300gsmWoven/UD	10m	External Impact Wrap
Epoxy	Clear Laminating (Low Vis cosity)	1 Gal	Composite Matrix
Polyolefin	3:1 Adhesive-Lined (Black)	10m	Lower Frame Armor
PET Bottles	1L Polyethylene	20+	DIY Heat-Shrink Consolida tion

Last Updated: Round 44 (Integration of sensing logic and pole framematerials)

